

Algebra 1 – Q1.4_1.7 B

Name: _____

1.4 – Fractions, Decimals & Percents

1. I can convert between fractions, decimals, and percents

Convert as directed. Write answers in simplest form.

- a. $\frac{5}{8} \rightarrow$ decimal: _____ percent: _____
- b. $0.45 \rightarrow$ fraction: _____ percent: _____
- c. $32\% \rightarrow$ fraction: _____ decimal: _____
- d. $0.875 \rightarrow$ fraction: _____ percent: _____

2. I can decide if a fraction's decimal ends or repeats (and write it).

For each fraction: **write the decimal** (use a bar for repeating digits).

- a. $\frac{13}{40} \rightarrow$ decimal: _____
- b. $\frac{2}{9} \rightarrow$ decimal: _____
- c. $\frac{5}{24} \rightarrow$ decimal: _____

3. I can compare and order fractions, decimals, and percents in real-world situations

Three stores are discounting headphones originally priced at **\$180**:

- Store A: **\$60 off**
- Store B: **35% off**
- Store C: **" $\frac{1}{2}$ off"**

Which store gives the **biggest discount**? _____

1.5 – Multiply, Divide & Compare Fractions

4. I can multiply fractions and mixed numbers

Simplify your answers.

a. $\frac{4}{7} \times 14 = \underline{\hspace{2cm}}$

b. $\frac{5}{6} \times \frac{9}{10} = \underline{\hspace{2cm}}$

c. $2\frac{1}{4} \times 1\frac{2}{3} = \underline{\hspace{2cm}}$

5. I can divide fractions by using reciprocals

Find the following quotients.

a. $\frac{5}{6} \div \frac{3}{8} = \underline{\hspace{2cm}}$

b. $7 \div \frac{3}{5} = \underline{\hspace{2cm}}$

1.6 – Solving Problems with Fractions, Decimals & Percents

6. I can find a part, a whole, or a percent

- a. What is 22% of 150? _____
- b. 28 is 35% of what number? _____
- c. What percent is 18 out of 24? _____

7. I can use percents to solve problems involving grades, sales, and survey results

- a. A \$60 pair of shoes is 15% off. After the discount, 8% sales tax is added. What is the final price? (Round to the nearest cent.)

Answer: _____

- b. In a survey of 200 students, 37% prefer pizza. How many students is that? _____

8. I can solve percent change problems in real-world contexts

A recipe uses $\frac{2}{3}$ cup of sugar. You **increase** the sugar by 25%. How much sugar do you use now?

Answer: _____ cup(s)

1.7 – Order of Operations (PEMDAS)

9. I follow the correct order of operations (including fractions and negatives)

Simplify each expression.

a. $5 - \frac{2}{3} \cdot 6 =$ _____

b. $-4^2 + 7 =$ _____

c. $\frac{3}{4} + \frac{1}{2} \cdot 3 - 2 =$ _____

d. $\frac{7}{2} \div \left(\frac{7}{4}\right) + 1 =$ _____

10. I avoid common mistakes when simplifying expressions

A student says $6 \div \frac{1}{3} = 2$ “because you divide by 3.”

a. Explain what is wrong.

b. Find the correct value of $6 \div \frac{1}{3}$.

11. I avoid common mistakes with negatives and exponents

True or False? Explain your reasoning clearly.

“ $(-4)^2 = -16$ ”